

Practice Problems (Set 7)

Intro to Probability and Statistics EE 354 / CE 361 / MATH 310

Question 1

X and Y are random variables with a joint PDF given by:

$$f_{X,Y}(x,y) = \frac{2}{c}$$

for $0 \leq x \leq 1$ and $0 \leq y \leq 1$. The joint PDF is zero for all other values of x and y.

- Find the value of c.
- Calculate the marginal PDFs of X and Y.

Question 2

W and X are independent continuous random variables that are uniformly distributed with $W \sim U[0,1]$ and $X \sim U[0,2]$. Find the Cumulative Distribution Function, CDF, and Probability Density Function, PDF, for the following random variables:

- $Y = \max(W, X)$
- $Z = \max(W, X) + 1$

Question 3

Let X and Y be two independent continuous random variables with uniform PDFs:

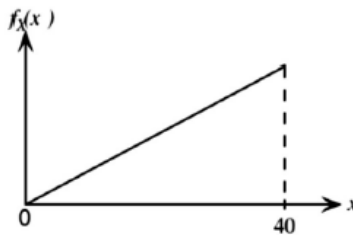
$$X \sim U[0,1]$$

$$Y \sim U[0,2]$$

- Find $P(\max(X, Y) \leq 0.8)$
- Find $f_{X,Y}(x, y)$

Question 4

A random variable X has the following PDF:



Conditional on random variable $X = x$, the continuous random variable Y is uniformly distributed between 0 and $3x$. Determine the joint PDF $f_{X,Y}(x, y)$ for $0 < x < 40$ and $0 < y < 3x$

Question 5

Let X be a standard Normal random variable i.e. $X \sim N(0,1)$. Let Y be an independent Bernoulli random variable taking on values 0 or 1 with equal probability, $p=1/2$. Also, consider the random variable $Z = X + Y$.

- Find the conditional PDF of X given $Y=1$.

- b) Find the conditional PDF of Z given Y=1.

Question 6

On a sunny day, it takes a student between 30 to 45 minutes to get to Habib University campus with all times being equally likely. On a rainy day, it takes the same student between 40 to 60 minutes to get to campus with all times being equally likely. Assume a day is sunny with probability 0.75 and rainy with probability 0.25.

- Find the PDF of the time T that it takes the student to get to campus.
- On a given day, it took the student 42 minutes to get to campus. What is the probability that this particular day was rainy?

Question 7

A binary signal S is transmitted and we are given that $P(S = 1) = 0.5$ and $P(S = -1) = 0.5$. The received signal is $Y = N + S$, where N is normal noise with zero mean and unit variance, independent of S. What is the probability that $S = 1$ if the observed value of received signal Y is 0.5.

Question 8

Smartphones produced by a particular manufacturer are known to have an exponentially distributed lifetime Y. However, the manufacturing plant has had some quality control problems lately. As a result, on any given day, the parameter λ of PDF of Y is actually a random variable uniformly distributed in the interval [1,1.5]. We decide to test a smartphone and record its lifetime. If the lifetime turns out to be 3 years, what is the updated PDF of the underlying parameter λ .

SOLUTIONS:

Question 1

$$i) \quad f_{X,Y}(x,y) = \begin{cases} \frac{2}{c} & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx dy = 1$$

$$\Rightarrow f_{X,Y}(x,y) = \begin{cases} 1 & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow c = 2$$

ii)

$$f_x(x) = \int_{-\infty}^{\infty} f_{x,y}(x,y) \cdot dy$$

$$= \int_0^1 1 \cdot dy = y \Big|_0^1$$

$$f_x(x) = 1$$

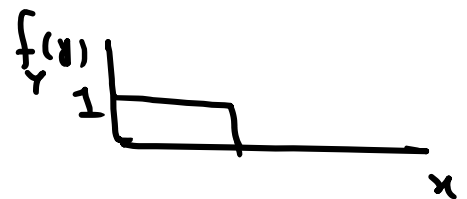
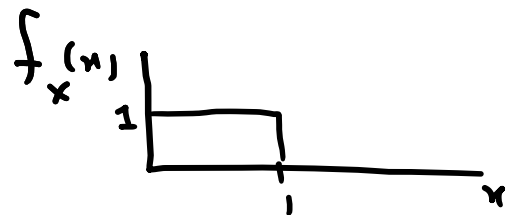
$$0 \leq x \leq 1$$

$$f_y(y) = \int_{-\infty}^{\infty} f_{x,y}(x,y) \cdot dx$$

$$= \int_0^1 1 \cdot dx = x \Big|_0^1$$

$$f_y(y) = 1$$

$$0 \leq y \leq 1$$

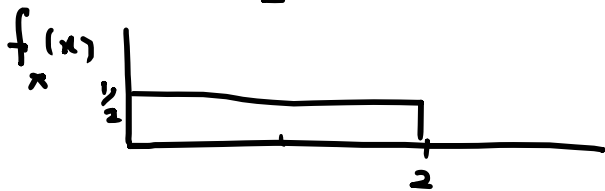
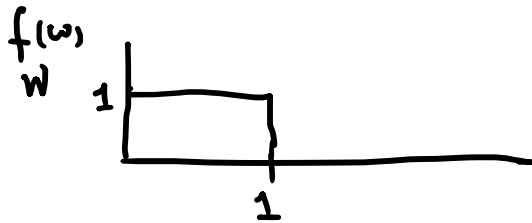


Question 2

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \stackrel{(a)}{=} P(\max(W, X) \leq y) \\ &= P(W \leq y, X \leq y) \end{aligned}$$

$\therefore W$ & X are independent

$$F_Y(y) = P(W \leq y) \cdot P(X \leq y) \quad \text{--- (1)}$$



Now

$$P(W \leq y) = \begin{cases} 0 & y < 0 \\ y & 0 \leq y < 1 \\ 1 & 1 \leq y \end{cases}$$

$$P(X \leq y) = \begin{cases} 0 & y < 0 \\ y/2 & 0 \leq y < 2 \\ 1 & 2 \leq y \end{cases}$$

From (1),

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ y/2 & 0 \leq y < 1 \\ y/2 & 1 \leq y < 2 \\ 1 & 2 \leq y \end{cases}$$

$$\therefore f_Y(y) = \frac{d}{dy} [F_Y(y)]$$

$$f_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 \leq y < 1 \\ \frac{1}{2} & 1 \leq y < 2 \\ 0 & 2 \leq y \end{cases}$$

(b)

Note that

$$Z = Y + 1$$

$\Rightarrow F_Z(z)$ and $f_Z(z)$ are just shifted (by 1) versions of $F_Y(y)$ and $f_Y(y)$

Question 3

$$P(\max(X, Y) \leq 0.8)$$

$$= P(X \leq 0.8, Y \leq 0.8)$$

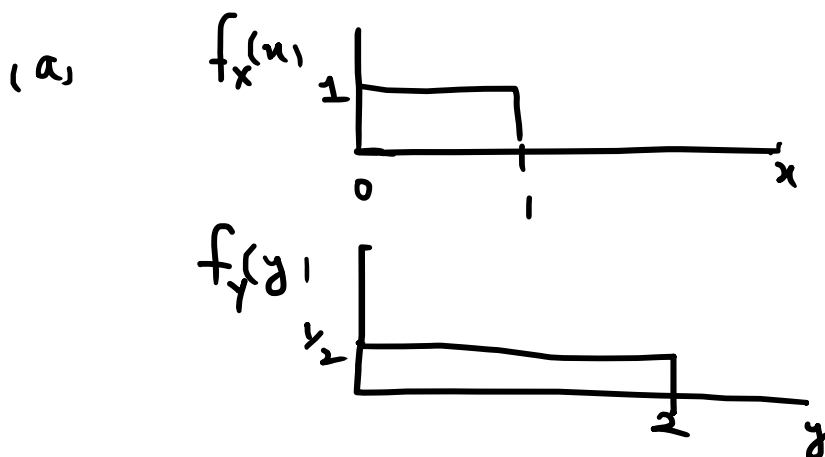
By Independence:

$$= P(X \leq 0.8) \cdot P(Y \leq 0.8)$$

$$= 1(0.8) \cdot \frac{1}{2}(0.8)$$

$$= 0.32$$

— Ans



(b)

By Independence,

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$f_{X,Y}(x,y) = \begin{cases} 1 \cdot \frac{1}{2} & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 \cdot \frac{1}{2} & 1 < x \leq 2, 0 \leq y \leq 2 \\ 0.0 & \text{otherwise} \end{cases}$$

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Question 4

$$f_{x,y}(x,y) = f_x(x) \cdot f_{y|x}(y|x)$$

Now,

$$f_x(x) = mx \quad 0 \leq x \leq 40$$

when

$$\Rightarrow f_x(x) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1}{20} - 0}{40 - 0} = \frac{1}{800} \quad \text{for } 0 \leq x \leq 40$$

— (1),

Also,

$$f_{y|x}(y|x) = U(0, 3x)$$

$$= \frac{1}{3x} \quad 0 \leq y \leq 3x$$

— (2),

Using (1),

$$f_{x,y}(x,y) = \begin{cases} \frac{1}{800}x \cdot \frac{1}{3x} = \frac{1}{2400} & 0 \leq x \leq 40 \\ & 0 \leq y \leq 3x \\ 0 & \text{otherwise} \end{cases}$$

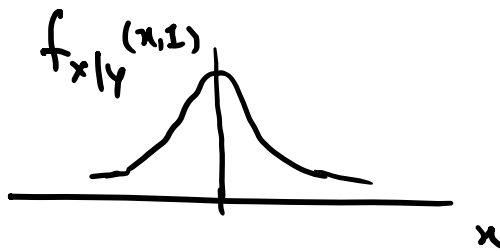
— Ans

Question 5

By Independence,

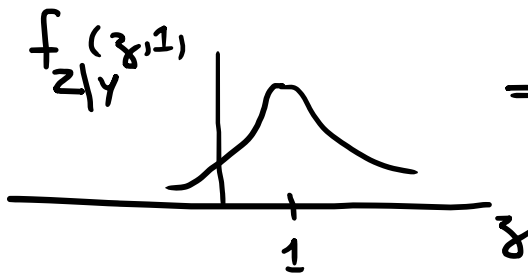
(a)

$$f_{X|Y}(x, y) \rightarrow f_X(x)$$



$$\Rightarrow f_{X|Y}(x, 1) = N(0, 1)$$

(b)



$$\Rightarrow f_{Z|Y}(z, 1) = N(1, 1)$$

Question 6

(a)

Let X be the random variable that takes the value 0 when it's a sunny day and value 1 when it's a rainy day.

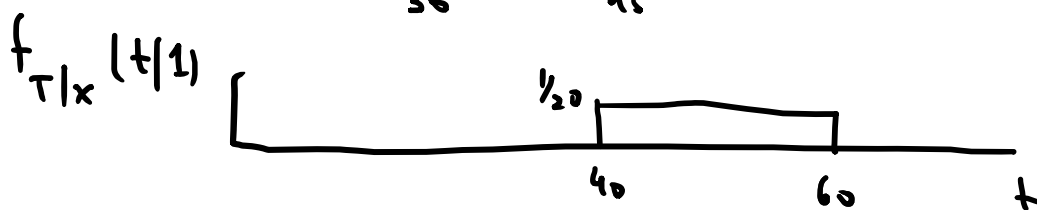
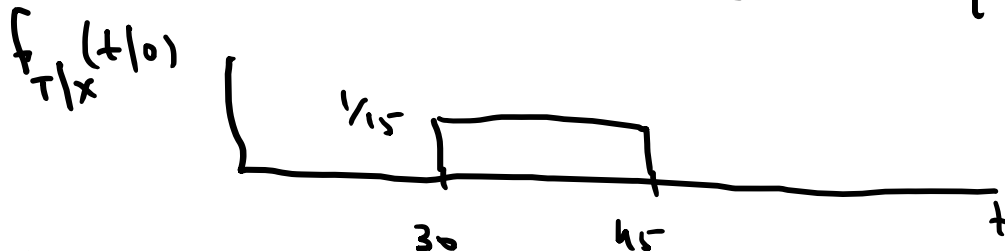
$$p_X(x) = \begin{cases} 0.75 & \text{when } x=0 \\ 0.25 & \text{when } x=1 \end{cases}$$

Now,

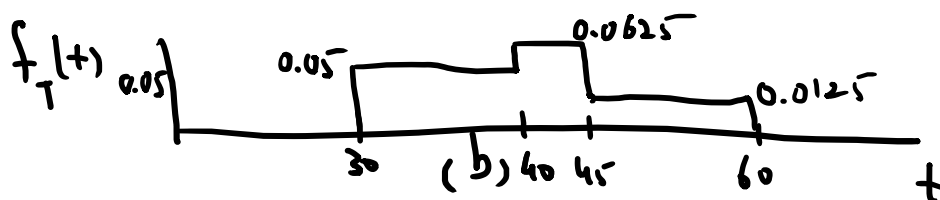
$$f_T(t) = \sum_x p_X(x) f_{T|X}(t|x)$$

$$= p_X(0) f_{T|X}(t|0) + p_X(1) f_{T|X}(t|1)$$

$$f_T(t) = 0.75 [U(30, 45)] + 0.25 [U(40, 60)] \quad \text{--- (1)}$$



From 1,



$$p_{X|T}(1|42) = ?$$

Using Bayes' Rule:- (Discrete X , Continuous Y)

$$p_{X|Y}(x|y) = \frac{p_X(x) f_{Y|X}(y|x)}{f_Y(y)}$$

$$\Rightarrow p_{X|T}(x|t) = \frac{p_X(x) f_{T|X}(t|x)}{f_T(t)}$$

$$p_{X|T}(1|42) = \frac{p_X(1) f_{T|X}(42|1)}{f_T(42)} \quad \text{--- (2)}$$

$$p_X(1) = 0.25$$

$$f_{T|X}(42|1) = \frac{1}{20} = 0.05$$

$$f_T(42) = 0.0625$$

From (2)

$$p_{X|T}(1|42) = \frac{0.25 \times 0.05}{0.0625}$$

$$p_{X|T}(1|42) = 0.2 \quad \text{--- Ans}$$

Question 7

Let S be the random variable representing the transmitted signal and Y be the random variable representing the received signal.

$$p_S(s) = \begin{cases} 0.5 & \text{when } s=1 \\ 0.5 & \text{when } s=-1 \end{cases}$$

$$f_{Y|S}(y|1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2}}$$

$$f_{Y|S}(y|-1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y+1)^2}{2}}$$

By Total Probability Theorem:

$$f_Y(y) = \sum_s p_S(s) f_{Y|S}(y|s)$$

$$= p_S(1) f_{Y|S}(y|1) + p_S(-1) f_{Y|S}(y|-1)$$

$$f_Y(y) = \frac{0.5}{\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2}} + \frac{0.5}{\sqrt{2\pi}} e^{-\frac{(y+1)^2}{2}}$$

$$p_{S|Y}(1|0.5) = ?$$

Using Bayes' Rule:

$$p_{S|Y}(s|y) = \frac{p_S(s) f_{Y|S}(y|s)}{f_Y(y)}$$

$$\begin{aligned}
 P_{S|Y}(1|0.5) &= \frac{P_S(1) f_{Y|S}(0.5|1)}{f_Y(0.5)} \\
 &= \frac{0.5 \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{(0.5-1)^2}{2}} \right)}{\frac{0.5}{\sqrt{2\pi}} e^{-\frac{(0.5-1)^2}{2}} + \frac{0.5}{\sqrt{2\pi}} e^{-\frac{(0.5+1)^2}{2}}} \\
 &= \frac{e^{-0.125}}{e^{-0.125} + e^{-1.125}} \\
 &= \frac{0.8825}{0.8825 + 0.3247}
 \end{aligned}$$

$$P_{S|Y}(1|0.5) = 0.731 \quad \text{--- Ans}$$

Question 8

$$f_{\lambda}(\lambda) = \begin{cases} 2 & 1 \leq \lambda \leq 1.5 \\ 0 & \text{otherwise} \end{cases} \quad \text{--- (1)}$$

$$f_{Y|\lambda}(y|\lambda) = \lambda e^{-\lambda y} \quad y \geq 0 \quad \text{--- (2)}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{\lambda}(t) f_{Y|\lambda}(y|t) dt$$

$$f_Y(y) = \int_1^{1.5} 2t e^{-ty} dt \quad \text{--- (3)}$$

$$f_{\lambda|Y}(\lambda|3) = ?$$

Using Bayes' Rule (Continuous X, Continuous Y):-

$$f_{\lambda|Y}(\lambda|y) = \frac{f_{\lambda}(\lambda) f_{Y|\lambda}(y|\lambda)}{f_Y(y)}$$

$$f_{\lambda|Y}(\lambda|3) = \frac{f_{\lambda}(\lambda) f_{Y|\lambda}(3|\lambda)}{f_Y(3)}$$

$$f_{\lambda|y}(\lambda|3) = \frac{2 \lambda e^{-3\lambda}}{\int_1^{1.5} 2t e^{-3t} \cdot dt}$$

$$1 \leq \lambda \leq 1.5$$

— Any